1.4 Four Particular Solutions

The equation $dy/dt = ay$ is solved by $y(t) = e^{at}y(0)$. All the input is in that starting value $y(0)$. The solution grows exponentially when $a > 0$ and it decays when $a < 0$. *This section allows new inputs $q(t)$ after the starting time $t = 0$.* That input q is a "source" when we add to $y(t)$, and a "sink" when we subtract. If $y(t)$ is the balance in a bank account at time t, then $q(t)$ is the rate of new deposits and withdrawals.

The basic first order linear differential equation (1) is fundamental to this course. We must and will solve this equation. Please pay attention to this section. In every way, this Section 1.4 is important.

$$\frac{dy}{dt} = ay + q(t) \qquad \textbf{starting from } y(0) \textbf{ at } t = 0. \tag{1}$$

Important **I will separate the solution $y(t)$ into two parts.** One part comes from the starting value $y(0)$. The other part comes from the source term $q(t)$. This separation is a crucial step for all linear equations, and I take this chance to give names to the two parts. The part $y_n = Ce^{at}$ is what we already know. The part y_p from the source $q(t)$ is new.

1 Homogeneous solution or null solution $y_n(t)$ with no source: $q = 0$

This part $y_n(t) = Ce^{at}$ solves the equation $dy/dt = ay$. The source term q is zero (null). We are really solving $y' - ay = 0$, an equation with zero on the right hand side. That equation is **homogeneous**—we can multiply a solution by any constant to get another solution $cy(t)$. This book will choose the simpler word *null* and the subscript n, because this connects differential equations to linear algebra.

2 Particular solution $y_p(t)$ with source $q(t)$

This part $y_p(t)$ comes from the source term $q(t)$. The previous section had no source and therefore no reason to mention $y_p(t)$. Now our whole task is to find a particular solution $y_p(t)$, because the null solutions $y_n(t) = Ce^{at}$ are already set.

3 The complete solution is $y(t) = y_n(t) + y_p(t)$

For linear equations—and only for linear equations—adding the two parts gives the complete solution $y = y_n + y_p$. This is also called the "general solution."

Null	y_n'	$-\ ay_n$	$=$	$\mathbf{0}$	y_n can start from $y(0)$
Particular	y_p'	$-\ ay_p$	$=$	$\boldsymbol{q(t)}$	y_p can start from $y = 0$
$\boldsymbol{y = y_n + y_p}$	$\boldsymbol{y'}$	$\boldsymbol{-\ ay}$	$=$	$\boldsymbol{q(t)}$	y must start from $y(0)$

A nonlinear equation could include a quadratic term y^2. In that case adding $y_n{}^2$ to $y_p{}^2$ would not give $(y_n + y_p)^2$. The null equation $y' - y^2 = 0$ would not be homogeneous, and we can't multiply y by a constant C. This will happen for the "logistic equation" in Section 1.7. You will see that $y(0)$ enters the solution $y(t)$ in a more complicated way.

The back cover of this book shows one particular solution y_p combining with all null solutions y_n. This important picture is repeated for matrix equations and linear algebra.

Particular Solutions and the Complete Solution

We can draw the complete solution to $u + v = 6$. These points (u, v) fill a straight line. We can also draw all the null solutions to $u + v = 0$. They fill a parallel straight line, going through the center point $(0, 0)$. Figure 1.2 shows how the null solutions combine with one particular solution $(3, 3)$ to give the line of complete solutions.

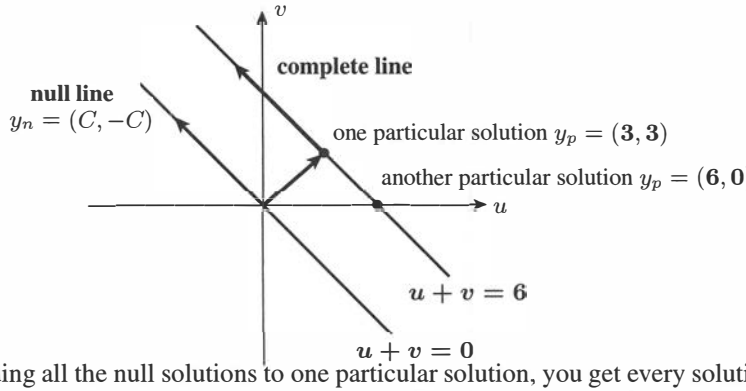


Figure 1.2: By adding all the null solutions to one particular solution, you get every solution (the complete line). You can start from *any* particular y_p that solves $u + v = 6$.

Starting from $y_p = (3, 3)$, the complete solution has $u = 3 + C$ and $v = 3 - C$. This includes a **null** solution $C + (-C) = 0$, plus the **particular** solution $3 + 3 = 6$.

Null	u_n	+	v_n	=	**0**	C	+	$(-C)$	=	0
Particular	u_p	+	v_p	=	**6**	3	+	3	=	6
Complete	u	+	v	=	**6**	$\boldsymbol{(3 + C)}$	+	$\boldsymbol{(3 - C)}$	=	**6**

The null solution $(C, -C)$ allows any constant C (like $y(0)$). The particular solution could have any numbers u_p and v_p that add to 6. We made a special choice $u_p = 3$ and $v_p = 3$. In the equation $y' - ay = q$ we will often make the special choice $y_p(0) = 0$.

There are many particular solutions! You could say that we chose a *very particular* solution. In the differential equation we chose to start from $y_p(0) = 0$. For the equation $u + v = 6$ we chose $u = 3$ and $v = 3$. *We could equally well choose* $u = 6$ *and* $v = 0$. This particular solution is different, but we get the same complete solution line:

$$y_{\text{complete}} = (6 + c, 0 - c) \text{ is the same solution line as } y_{\text{complete}} = (3 + C, 3 - C).$$

If c is 5, then C is 8. From all c's and all C's, you get the same line.

I want to repeat this pattern of *null solution plus particular solution* by showing how it looks for an ordinary matrix equation $A\boldsymbol{v} = \boldsymbol{b}$ (Chapter 4 explains matrices):

Null solution $A\boldsymbol{v}_n = \boldsymbol{0}$ **Particular solution** $A\boldsymbol{v}_p = \boldsymbol{b}$ **Complete solution** $\boldsymbol{v} = \boldsymbol{v}_n + \boldsymbol{v}_p$

Always the key is *linearity*: $A\boldsymbol{v}$ *equals* $A\boldsymbol{v}_n + A\boldsymbol{v}_p$. Therefore $A\boldsymbol{v} = \boldsymbol{0} + \boldsymbol{b} = \boldsymbol{b}$.

Often the only solution to $A\boldsymbol{v}_n = \boldsymbol{0}$ is $\boldsymbol{v}_n = \boldsymbol{0}$. Then a particular solution $\boldsymbol{v}_p$ is also the complete solution. This will happen when A is an "invertible matrix."

Inputs $q(t)$ and Responses $y(t)$

For any input source $q(t)$, equation (4) will solve $dy/dt = ay + q(t)$. But when mathematics is applied to science and engineering and our society, problems don't involve "any $q(t)$." *Certain functions $q(t)$ are the most important.* Those functions are constantly met in applied mathematics. Here is a short list of special inputs:

1. **Constant source** $q(t) = q$
2. **Step function at T** $q(t) = H(t - T)$
3. **Delta function at T** $q(t) = \delta(t - T)$
4. **Exponential** $q(t) = e^{ct}$

This section will solve $dy/dt = ay + q(t)$ for the four functions on that short list. The next section adds one more source $q(t)$. It is a combination of sine and cosine. Or $q(t)$ can be a complex exponential (which has one term and is usually easier):

5. **Sinusoid** $q(t) = A\cos\omega t + B\sin\omega t$ or $Re^{i\omega t}$

Solving Linear Equations by an Integrating Factor

The equation $y' = ay + q$ is so important that I will solve it in different ways. The first way uses an integrating factor $M(t)$. Put both y terms on the left. Keep $q(t)$ on the right.

Problem **Solve $y' - ay = q(t)$ starting from any $y(0)$**

Method ***Multiply both sides by the integrating factor $M(t) = e^{-at}$.***

We chose that factor e^{-at} so that M times $y' - ay$ is exactly the derivative of My:

$$\textbf{Perfect derivative} \qquad e^{-at}(y' - ay) \quad \text{agrees with} \quad \frac{d}{dt}(e^{-at}y) = \frac{d}{dt}(My). \tag{2}$$

When both sides of $y' - ay = q$ are multiplied by $M = e^{-at}$, our equation is immediately ready to be integrated. The right side is Mq, the left side is the derivative of My.

$$\text{The integral of } \frac{d}{dt}(My) = Mq \quad \text{is} \quad M(t)\,y(t) - M(0)\,y(0) = \int_0^t M(s)\,q(s)\,ds \tag{3}$$

At $t = 0$ we know that $M(0) = e^0 = 1$. Multiply both sides of equation (3) by e^{at} (which is $1/M$) to see $y(t) = y_n + y_p$. This solution comes many times in the book! To give meaning to formula (4), I will apply it to the most important inputs $q(t)$.

The key formula
Solution to $y' = ay + q(t)$

$$y(t) = e^{at}\,y(0) + e^{at}\int_0^t e^{-as}\,q(s)\,ds. \tag{4}$$

Constant Source $q(t) = q$

When $q(t)$ is a constant, the integration for the particular solution in equation (4) is easy.

$$\int_0^t e^{-as}\, q\, ds = \left[\frac{qe^{-as}}{-a}\right]_{s=0}^{s=t} = \frac{q}{a}(1 - e^{-at}).$$

Multiply by e^{at} to find $y_p(t)$. An important solution to an important equation.

Solution for constant source q $$\boldsymbol{y(t) = e^{at}\, y(0) + \frac{q}{a}(e^{at} - 1)} \qquad (5)$$

Example 1 has a positive growth rate $a > 0$. The solution will increase when $q > 0$. Example 2 will have a *negative* rate $a < 0$. In that case $y(t)$ approaches a *steady state*.

Example 1 Solve $dy/dt - 5y = 3$ starting from $y(0) = 2$. Here $a = 5$ and $q = 3$.

This fits perfectly with $y' - ay = q$. Equation (5) gives the solution $y(t)$:

Solution $y(t) = y_n + y_p = \mathbf{2e^{5t} + \frac{3}{5}(e^{5t} - 1)}$. Set $t = 0$ to check that $y(0) = 2$.

Looking at that solution, I have to admit that $y' - 5y = 3$ is not so obvious. This becomes much clearer when the two parts (null + particular) are separated:

$$y_n(t) = 2e^{5t} \text{ certainly has } y_n' - 5y_n = 0 \text{ with } y_n(0) = 2$$
$$y_p(t) = \tfrac{3}{5}(e^{5t} - 1) \text{ has } y_p' = 3e^{5t}. \text{ This agrees with } 5y_p + 3.$$

Example 2 Solve $dy/dt = 3 - 6y$ starting from $y(0) = 2$.

Formula (5) still gives the answer, but this $y(t)$ is decreasing because $\boldsymbol{a = -6}$ **is negative**:

$$y(t) = 2e^{-6t} + \frac{3}{-6}(e^{-6t} - 1) = \mathbf{\frac{3}{2}e^{-6t} + \frac{1}{2}}.$$

When $t = 0$, that solution starts at $y(0) = 2$. The solution decreases because of e^{-6t}. **As $\boldsymbol{t \to \infty}$ the solution approaches $\boldsymbol{y_\infty = \frac{1}{2}}$.** This value $-q/a$ at $t = \infty$ is a *steady state*.

At $y = -\dfrac{q}{a} = \dfrac{1}{2}$ the equation $\dfrac{dy}{dt} = 3 - 6y$ becomes $\boldsymbol{\dfrac{dy}{dt} = 0}$. Nothing moves.

Please notice that the steady state is $y_\infty = \frac{1}{2}$ for every initial value $y(0)$. That is because the null solution $y_n = y(0)e^{-6t}$ approaches zero. It is the particular solution that balances the source term $q = 3$ with the decay term $ay = -6y$ to approach $\boldsymbol{y_\infty = -q/a = 3/6}$.

Question If $y(0) = \frac{1}{2}$, what is $y(t)$? *Answer* $\boldsymbol{y(t) = \frac{1}{2}}$ **at all times**. $6y$ balances 3.

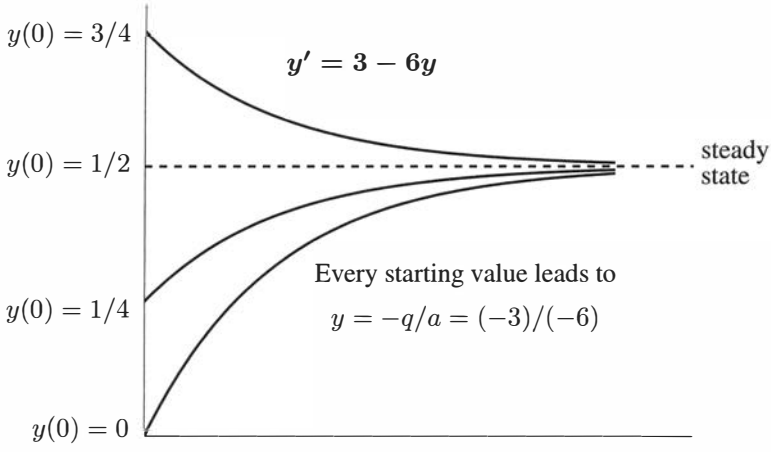


Figure 1.3: When a is negative, e^{at} approaches zero and $y(t)$ approaches $y_\infty = -q/a$.

Here is an important way to rewrite that basic equation $y' = ay + q$ when $a < 0$. The right hand side is the same as $a(y + \frac{q}{a})$. But $\boldsymbol{y + \frac{q}{a}}$ is exactly the distance $\boldsymbol{y - y_\infty}$. **Rewrite $\boldsymbol{y' = ay + q}$ as an easy equation $\boldsymbol{Y' = aY}$ by introducing $\boldsymbol{Y = y - y_\infty}$.**

New unknown $Y = y - y_\infty$ **New equation** $\boldsymbol{Y' = aY}$ **New start** $Y(0) = y(0) - y_\infty$

The solution to $Y' = aY$ is certainly $Y(t) = Y(0)e^{at}$. This approaches $Y_\infty = 0$ when $a < 0$. The original $y = Y + y_\infty$ still approaches y_∞ which is $-q/a$: see Figure 1.3.

$$\boldsymbol{(y - y_\infty)' = a(y - y_\infty)} \quad \textbf{has solution} \quad \boldsymbol{y(t) - y_\infty = e^{at}(y(0) - y_\infty)} \tag{6}$$

Section 1.6 will present physical examples with $a < 0$: Newton's Law of Cooling, the level of messenger RNA, the decaying concentration of a drug in the bloodstream.

Step Function

The unit step function or "Heaviside step function" $\boldsymbol{H(t)}$ jumps from 0 to 1 at $\boldsymbol{t = 0}$. Figure 1.4 shows its graph. The effect of $H(t)$ is like turning on a switch.

The second graph shows a *shifted step function* $H(t - T)$ which jumps from 0 to 1 at time T. This is the moment when $t - T = 0$, so H jumps at that moment T.

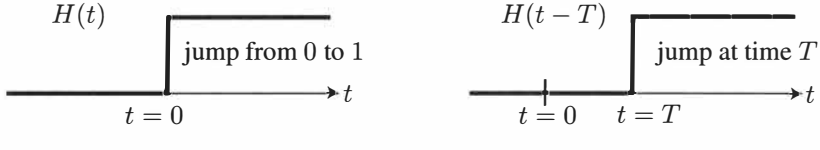


Figure 1.4: **The unit step function is $\boldsymbol{H(t)}$.** Its shift $H(t - T)$ jumps to 1 at $t = T$.

When the step comes at $t = 0$, the solution to $y' - ay = H(t)$ is the step response. That step response is easy to find because this equation is simply $y' - ay = 1$. The starting value is $y(0) = 0$. Put $q = 1$ into formula (5):

Step response $$y(t) = \frac{1}{a}(e^{at} - 1) \tag{7}$$

The interesting case is $a < 0$. The solution starts at $y(0) = 0$. It grows to $y(\infty) = -1/a$. The system rises to that steady state after the switch is turned on. The graph of $y(t)$ is the bottom curve in Figure 1.3, except that y_∞ is $1/6$ because the step function has $q = 1$.

The step response is the output $y(t)$ when the step function is the input. We are depositing at a constant rate $q = 1$. But when $a < 0$, we are losing ay in real value because of inflation. Then growth stops at $y = -1/a$, where the deposits just balance the loss.

Now turn on the switch at time T instead of time 0. The step function $H(t - T)$ is piecewise constant with two pieces: zero and one. If I multiply by any constant q, the source $q\,H(t - T)$ jumps from 0 to strength q at time T.

The left side of our differential equation is still $y' - ay$, no change. The integrating factor $M = e^{-at}$ still makes that into a perfect derivative: $M(y' - ay)$ *equals* $(My)'$. The only change is on the right side, where the constant source doesn't start acting until the jump time T. At that time, the step function source $H(t - T)$ is turned on:

$$(e^{-at}y)' = e^{-at}\,H(t-T) \text{ now gives } e^{-at}y(t) - e^{0t}\,y(0) = \int_T^t e^{-as}\,ds. \tag{8}$$

The only change for $t \geq T$ is to **start that integral at the turn-on time T**:

$$\int_T^t e^{-as}\,ds = \left[\frac{e^{-as}}{-a}\right]_{s=T}^{s=t} = \frac{1}{a}(e^{-aT} - e^{-at}). \tag{9}$$

Multiply by e^{at} to get the particular solution $y_p(t)$ beyond time T, and add $y_n = e^{at}y(0)$.

Solution with unit step $$y(t) = e^{at}y(0) + \frac{1}{a}\,(e^{a(t-T)} - 1) \text{ for } t \geq T. \tag{10}$$

As always, $y(0)$ grows or decays with e^{at} in the null solution y_n. The step response is the particular solution, as soon as the input begins. But nothing enters until time T.

Example 3 Suppose the input turns on at time $t = 0$ and turns off at $t = T$. Find $y(t)$.

Solution The input is $H(t) - H(t-T)$. The output is $y(t) = \frac{1}{a}\left(e^{at} - e^{a(t-T)}\right), t \geq T$.

Delta Function

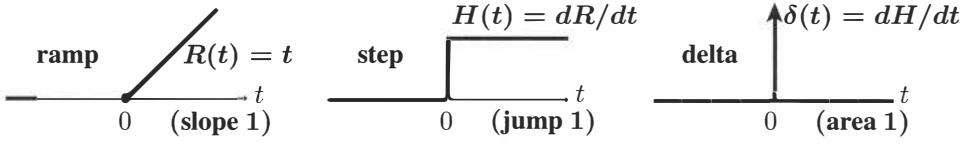
Now we meet a remarkable function $\delta(t)$. This "delta function" is everywhere zero, except at the instant $t = 0$. In that one moment it gives a unit input. Instead of a continuing source spread out over time, $\delta(t)$ is a **point source** completely concentrated at $t = 0$.

For a point source shifted to $\delta(t - T)$, **everything enters exactly at time T.** There is no source before that time or after that time. The delta function is zero except at one point. This "**impulse** " is by no means an ordinary function.

Here is one way to think about $\delta(t)$. **The delta function is the derivative of the unit step function $H(t)$.** But H is constant and dH/dt is zero except at $t = 0$. Take the integral of $\delta(t) = dH/dt$ from any negative number N to any positive number P.

Integral of $\delta(t)$ is 1
$$\int_N^P \boldsymbol{\delta(t)}\, dt = \int_N^P \frac{\boldsymbol{dH}}{\boldsymbol{dt}}\, dt = H(P) - H(N) = \mathbf{1 - 0}. \qquad (11)$$

"The area under the graph of $\delta(t)$ is 1. All that area is above the single point $t = 0$." Those words are in quotes because area at a point is impossible for ordinary functions. $\delta(t)$ may seem new and strange (it is useful!). Look at $dR/dt = H$ and $dH/dt = \delta$.



Slope of the ramp jumps to 1. Slope of the step function is the delta function.

The value of $\delta(0)$ is infinite. But that one word does not give full information. **The real way to understand delta functions is by their integrals.**

$$\int_{-\infty}^{\infty} \delta(t)\, dt = \mathbf{1} \qquad \int_{-\infty}^{\infty} \delta(t)\, F(t)\, dt = \boldsymbol{F(0)} \qquad \int_{-\infty}^{\infty} \delta(t - T)\, F(t)\, dt = \boldsymbol{F(T)} \qquad (12)$$

Please visualize a tall thin box function—equal to $1/h$ between $t = 0$ and $t = h$. Now imagine h going to zero. The width h becomes zero and the height $1/h$ becomes infinite. *The area stays at* 1. All integrals of $\delta(t)F(t)$ are concentrated at $t = 0$: the "spike" .

Here is a quick way to solve $y' - ay = \delta(t)$, and then we will do it more slowly. We know that the derivative of a step function $H(t)$ is the delta function $\delta(t)$. So the derivative of the step response must be the impulse response:

$$\frac{\boldsymbol{d}}{\boldsymbol{dt}}\mathbf{(step)} = \mathbf{delta} \qquad \frac{\boldsymbol{d}}{\boldsymbol{dt}}\left(\begin{matrix}\mathbf{step}\\ \mathbf{response}\end{matrix}\right) = \frac{\boldsymbol{d}}{\boldsymbol{dt}}\left(\frac{\boldsymbol{e^{at}} - \mathbf{1}}{\boldsymbol{a}}\right) = \boldsymbol{e^{at}} = \begin{matrix}\mathbf{impulse}\\ \mathbf{response}\end{matrix} \qquad (13)$$

The Impulse Response Solves $y' - ay = \delta(t)$

Start your bank account with one deposit. Start your heart with a sudden shock. Hit a golf ball. Fire a bullet. Many motions start with an "impulse" and then the source term is a delta function $\delta(t)$.

The impulse response $y(t)$ jumps immediately to $y(0) = 1$. You can see that by integrating every term in $dy/dt - ay = \delta(t)$. Integrating $\delta(t)$ from $t = -h$ to h gives **1**. Integrating dy/dt gives $y(h) - y(-h)$, which is $\boldsymbol{y(h)}$. The integral of ay becomes zero as $h \to 0$. That limit step when $h \to 0$ leaves $\boldsymbol{y(0) = 1}$.

After the jump to $y(0) = 1$, the impulse $\delta(t)$ is immediately zero. So we just have the ordinary null solution to $y' = ay$ starting from $y(0) = 1$:

Impulse response $\quad y' - ay = \delta(t) \quad y(t) = e^{at}$ (14)

Notice the different responses to an impulse and a step function. The impulse deposits everything at $t = 0$. The step function goes on depositing forever. If $a < 0$ and inflation reduces our wealth, the impulse response dies out to $y_\infty = 0$. The step response increases from 0 to $y_\infty = -1/a$, where the deposits balance the loss from inflation.

I want to emphasize: **e^{at} is the growth or decay factor $G(t)$ for all inputs**. When the input is $y(0)$, the output at time t is $e^{at}y(0)$. When the input is $q(s)$ at time s, the output later at t is $e^{a(t-s)}q(s)$. The growth is only over the remaining time $t - s$. **Our main formula (4) is adding up all the outputs that come from all the inputs.**

Delayed Delta Function

The source $q(t) = \delta(t - T)$ turns on at time T. Then immediately it turns off. In that one instant of time, *the value of y jumps by* 1. "We deposited \$1 at that moment." The integral of $dy/dt = \delta(t - T)$ is 1. This is the change in y, before T to after T.

Coming up to time T, the solution is $y(t) = e^{at}y(0)$. At time T we add 1. After time T, that input has the shorter period $t - T$ in which to grow. Multiply 1 by $e^{a(t-T)}$:

Solution for $q = \delta(t - T)$ $\quad \boldsymbol{y(t) = y_n(t) + y_p(t) = e^{at}\,y(0) + e^{a(t-T)}}$. (15)

The solution y jumps by $e^{a(T-T)} = e^0 = 1$, when that second term appears at $t = T$.

Example 4 Solve the equation $\boldsymbol{y' - 5y = 3\delta(t - 4)}$ starting from $y(0) = 2$.
The null solution to $y' - 5y = 0$ starting at $y(0) = 2$ is $\boldsymbol{y_n(t) = 2e^{5t}}$. This we know. The particular solution is $y_p(t) = 0$ up to $t = 4$. At that moment y jumps by 3, from 3δ. Its growth factor is $e^{5(t-4)}$. Then $\boldsymbol{y_p(t) = 3e^{5(t-4)}}$ after $t = 4$.

Complete solution with jump of 3 $\quad \boldsymbol{y_n + y_p = 2e^{5t} + 3e^{5(t-4)}H(t - 4)}$ (16)

The step function $H(t - 4)$ combines $y_p = 0$ before the jump and y_p after the jump into one formula. At $t = 4$ the solution jumps by 3. Then this 3 grows to $3e^{5(t-4)}$.

***Remark* 1** This solution makes me realize that the initial value $y(0)$ is like having a delta function at time $t = 0$. *The solution "jumps" to* $y(0)$. I don't know if you agree with that.

***Remark* 2** $q(t) = -\delta(t - T)$ would be negative (a sink instead of a source). A bank account could be earning interest at the rate a, and suddenly you withdraw 1 at time T. The balance $y(T)$ had reached $e^{aT}y(0)$, and it drops by 1. From time T onwards, the growth factor $e^{a(t-T)}$ multiplies the new balance, and $y(t) = e^{at}y(0) - e^{a(t-T)}$.

***Remark* 3** (a little mysterious) We could think of an ordinary continuous input $q(t)$ as a lot of delta functions—a delta function of strength $q(T)$ *at every time* T. Instead of "a lot" I need to say "an integral". Every continuous function $q(t)$ is an integral of delta functions $q(T)\,\delta(t - T)$ at all T. The integral picks out $q(t)$ at the spike point.

$$\textbf{Any } \boldsymbol{q(t)} \textbf{ = combination of delta functions} = \int q(T)\,\delta(t - T)\,dT. \tag{17}$$

Example 5 ($q = 1$) **The integral of all impulses for $T \geq 0$ is the step function $H(t)$.**

Then the integral of all impulse responses is the step response. The integral of e^{at} from 0 to t is $(e^{at} - 1)/a$. *Derivative of step response = impulse response* as in (13).

Exponential Input e^{ct}

The source $q(t) = e^{ct}$ starts at time zero and continues forever. The particular solution $y_p(t)$ is easy to find, because $\boldsymbol{y_p}$ **is a multiple $\boldsymbol{Ye^{ct}}$ of this same exponential $\boldsymbol{e^{ct}}$.** That is the beauty of exponentials . These are the most important functions and the best to work with. They allow growth or decay or oscillation from $c > 0$ and $c < 0$ and $c = i\omega$.

Substitute $\boldsymbol{y_p = Ye^{ct}}$ into $\boldsymbol{y' - ay = e^{ct}}$ $\qquad cYe^{ct} - aYe^{ct} = e^{ct}$

When we cancel e^{ct} this leaves a simple formula for the number Y in Ye^{ct} :

$$\boldsymbol{cY - aY = 1} \quad \textbf{gives} \quad \boxed{Y = \frac{1}{c - a}} \quad \textbf{and} \quad \boxed{y_p(t) = \frac{e^{ct}}{c - a}} \tag{18}$$

Example 6 Solve $\boldsymbol{y' - 5y = 3e^{4t}}$ starting from $y(0) = 2$. Now $Y = \dfrac{3}{c - a} = \dfrac{3}{4 - 5}$.

The null solution still involves e^{5t}. The particular solution is Y times e^{4t} !

$$\boldsymbol{y_p(t) = Ye^{4t}} \qquad y_p' - 5y_p = (4Y - 5Y)e^{4t} = 3e^{4t}. \quad \textbf{Then } \boldsymbol{Y = -3}.$$

This particular solution $-3e^{4t}$ starts at -3. Since $y(0) = 2$, the other part starts at $+5$.

$$\textbf{Complete solution} \qquad \boldsymbol{y(t) = 5e^{5t} - 3e^{4t}}.$$

The null solution grows at rate $a = 5$. One particular solution grows at rate $c = 4$. The equation $y' - ay = e^{ct}$ is solved for $c \neq a$ but two final comments are needed.

1. This particular solution $y(t) = e^{ct}/(c - a)$ is not the "very particular" solution that starts from $y_p(0) = 0$. It is still perfectly good, except it starts at $1/(c - a)$. So the complete solution starting at $y(0)$ has to include the usual $y(0)e^{at}$ and **also a term to cancel $1/(c - a)$ at time zero**:

$$y' - ay = e^{ct} \qquad y_{\text{complete}} = y(0)\,e^{at} - \frac{e^{at}}{c-a} + \frac{e^{ct}}{c-a} \tag{19}$$

   There you see a null solution y_n (two terms) and our particular y_p (the last term). Or the last two terms together are the very particular solution $(e^{ct} - e^{at})/(c - a)$.

2. For $c = a$ we are in serious trouble. The formulas fail because we can't divide by $c - a = 0$. This problem $y' - ay = e^{at}$ is a type of **resonance**, when the exponent c in the source happens to equal the exponent a in the natural growth from $y' = ay$. The integral in our main formula (4) becomes $\int e^{-as}\, e^{as}\, ds = \int 1\, ds = t$.

$$\textbf{Resonance} \quad c = a \qquad y' - ay = e^{at} \qquad y = y(0)e^{at} + te^{at} \tag{20}$$

   That extra growth factor t is because y_n resonates with y_p. They both have e^{at}.

■ REVIEW OF THE KEY IDEAS ■

1. **Complete** solution to a linear equation = **null** solution(s) + **particular** solution.
2. The integrating factor e^{-at} multiplies $y' - ay = q(t)$ to give $(e^{-at}\, y)' = e^{-at}\, q(t)$.
   Integrate and multiply by e^{at}: $y(t) = y_n + y_p = e^{at}y(0) + e^{at} \int e^{-as} q(s)\, ds$.
3. For $y' - ay = q =$ **constant**, the particular solution with $y_p(0) = 0$ is $q(e^{at} - 1)/a$.
4. $q(t) = H(t)$: the response to a unit step function is $y_p = (e^{at} - 1)/a$.
5. $q(t) = \delta(t)$: the impulse response to a unit delta function is $y_p = e^{at}$.
6. $q(t) = e^{ct}$ gives $y_p = (e^{ct} - e^{at})/(c - a)$. In case $c = a$, change to $y_p = te^{at}$.

Problem Set 1.4

1 All solutions to $dy/dt = -y + 2$ approach the steady state where dy/dt is zero and $y = y_\infty =$ ___. That constant $y = y_\infty$ is a particular solution y_p.

Which $y_n = Ce^{-t}$ combines with this steady state y_p to start from $y(0) = 4$? This question chose $y_p + y_n$ to be $y_\infty +$ *transient* (decaying to zero).

2 For the same equation $dy/dt = -y + 2$, choose the null solution y_n that starts from $y(0) = 4$. Find the particular solution y_p that starts from $y(0) = 0$.

This splitting chooses the two parts $e^{at}y(0)$ + integral of $e^{a(t-s)}q$ in equation (4).

3 The equation $dy/dt = -2y + 8$ has two natural splittings $\boldsymbol{y_S + y_T = y_N + y_P}$:

1. Steady ($\boldsymbol{y_S = y_\infty}$) + Transient ($\boldsymbol{y_T \to 0}$). What are those parts if $y(0) = 6$?

2. ($y_N' = -2y_N$ from $\boldsymbol{y_N(0) = 6}$) + ($y_P' = -2y_P + 8$ starting from $\boldsymbol{y_P(0) = 0}$).

4 All null solutions to $u - 2v = 0$ have the form $(u, v) = (c,$ _____$)$.

One particular solution to $u - 2v = 3$ has the form $(u, v) = (7,$ _____$)$.

Every solution to $u - 2v = 3$ has the form $(7,$ _____$) + c(1,$ _____$)$.

But also every solution has the form $(3,$ _____$) + C(1,$ _____$)$ for $C = c + 4$.

5 The equation $dy/dt = 5$ with $y(0) = 2$ is solved by $y =$ _____. A natural splitting $y_n(t) =$ ___ and $y_p(t) =$ ___ comes from $y_n = e^{at}y(0)$ and $y_p = \int e^{a(t-s)}5\,ds$.

This small example has $\boldsymbol{a = 0}$ (so ay is absent) and $\boldsymbol{c = 0}$ (the source is $q = 5e^{0t}$). When $a = c$ we have "resonance." A factor t will appear in the solution y.

Starting with Problem 6, choose the very particular $\boldsymbol{y_p}$ that starts from $\boldsymbol{y_p(0) = 0}$.

6 For these equations starting at $y(0) = 1$, find $y_n(t)$ and $y_p(t)$ and $y(t) = y_n + y_p$.

(a) $y' - 9y = 90$ (b) $y' + 9y = 90$

7 Find a linear differential equation that produces $y_n(t) = e^{2t}$ and $y_p(t) = 5(e^{8t} - 1)$.

8 Find a resonant equation ($a = c$) that produces $y_n(t) = e^{2t}$ and $y_p(t) = 3te^{2t}$.

9 $y' = 3y + e^{3t}$ has $y_n = e^{3t}y(0)$. Find the resonant y_p with $y_p(0) = 0$.

Problems 10–13 are about $\boldsymbol{y' - ay =}$ constant source $\boldsymbol{q}$.

10 Solve these linear equations in the form $y = y_n + y_p$ with $y_n = y(0)e^{at}$.

(a) $y' - 4y = -8$ (b) $y' + 4y = 8$ Which one has a steady state?

11 Find a formula for $y(t)$ with $y(0) = 1$ and draw its graph. What is y_∞?

(a) $y' + 2y = 6$ (b) $y' + 2y = -6$

12 Write the equations in Problem 11 as $Y' = -2Y$ with $Y = y - y_\infty$. What is $Y(0)$?

13 If a drip feeds $q = 0.3$ grams per minute into your arm, and your body eliminates the drug at the rate $6y$ grams per minute, what is the steady state concentration y_∞? Then *in* = *out* and y_∞ is constant. Write a differential equation for $Y = y - y_\infty$.

Problems 14–18 are about $y' - ay =$ step function $H(t - T)$:

14 Why is y_∞ the same for $y' + y = H(t - 2)$ and $y' + y = H(t - 10)$?

15 Draw the ramp function that solves $y' = H(t - T)$ with $y(0) = 2$.

16 Find $y_n(t)$ and $y_p(t)$ as in equation (10), with step function inputs starting at $T = 4$.

(a) $y' - 5y = 3H(t - 4)$ (b) $y' + y = 7H(t - 4)$ (*What is* y_∞?)

17 Suppose the step function turns on at $T = 4$ and off at $T = 6$. Then $q(t) = H(t - 4) - H(t - 6)$. Starting from $y(0) = 0$, solve $y' + 2y = q(t)$. What is y_∞?

18 Suppose $y' = H(t - 1) + H(t - 2) + H(t - 3)$, starting at $y(0) = 0$. Find $y(t)$.

Problems 19–25 are about delta functions and solutions to $y' - ay = q\,\delta(t - T)$.

19 For all $t > 0$ find these integrals $a(t), b(t), c(t)$ of point sources and graph $b(t)$:

$$\text{(a)} \int_0^t \delta(T-2)\,dT \quad \text{(b)} \int_0^t (\delta(T-2) - \delta(T-3))\,dT \quad \text{(c)} \int_0^t \delta(T-2)\delta(T-3)dT$$

20 Why are these answers reasonable? (They are all correct.)

$$\text{(a)} \int_{-\infty}^{\infty} e^t\delta(t)dt = 1 \quad \text{(b)} \int_{-\infty}^{\infty} (\delta(t))^2 dt = \infty \quad \text{(c)} \int_{-\infty}^{\infty} e^T\delta(t-T)dT = e^t$$

21 *The solution to* $y' = 2y + \delta(t - 3)$ *jumps up by* 1 *at* $t = 3$. Before and after $t = 3$, the delta function is zero and y grows like e^{2t}. Draw the graph of $y(t)$ when (a) $y(0) = 0$ and (b) $y(0) = 1$. Write formulas for $y(t)$ before and after $t = 3$.

22 Solve these differential equations starting at $y(0) = 2$:

(a) $y' - y = \delta(t - 2)$ (b) $y' + y = \delta(t - 2)$. (*What is* y_∞?)

23 Solve $dy/dt = H(t - 1) + \delta(t - 1)$ starting from $y(0) = 0$: jump and ramp.

24 (My small favorite) What is the steady state y_∞ for $y' = -y + \delta(t - 1) + H(t - 3)$?

25 Which q and $y(0)$ in $y' - 3y = q(t)$ produce the step solution $y(t) = H(t - 1)$?

Problems 26–31 are about exponential sources $q(t) = Qe^{ct}$ and resonance.

26 Solve these equations $y' - ay = Qe^{ct}$ as in (19), starting from y(0) = 2 :

(a) $y' - y = 8e^{3t}$ (b) $y' + y = 8e^{-3t}$ (*What is* y_∞ ?)

27 When $c = 2.01$ is very close to $a = 2$, solve $y' - 2y = e^{ct}$ starting from $y(0) = 1$. By hand or by computer, draw the graph of $y(t)$: near resonance.

28 When $c = 2$ is exactly equal to $a = 2$, solve $y' - 2y = e^{2t}$ starting from $y(0) = 1$. This is resonance as in equation (20). By hand or computer, draw the graph of $y(t)$.

29 Solve $y' + 4y = 8e^{-4t} + 20$ starting from $y(0) = 0$. What is y_∞ ?

30 The solution to $y' - ay = e^{ct}$ didn't come from the main formula (4), but it could. Integrate $e^{-as}e^{cs}$ in (4) to reach the very particular solution $(e^{ct} - e^{at})/(c - a)$.

31 *The easiest possible equation* $y' = 1$ *has resonance* ! The solution $y = t$ shows the factor t. What number is the growth rate a and also the exponent c in the source ?

32 Suppose you know two solutions y_1 and y_2 to the equation $y' - a(t)y = q(t)$.

(a) Find a null solution to $y' - a(t)y = 0$.

(b) Find all null solutions y_n. Find all particular solutions y_p.

33 Turn back to the first page of this Section 1.4. Without looking, can you write down a solution to $y' - ay = q(t)$ for all four source functions $\boldsymbol{q}, \boldsymbol{H(t)}, \boldsymbol{\delta(t)}, \boldsymbol{e^{ct}}$?

34 Three of those sources in Problem 33 are actually the same, if you choose the right values for q and c and $y(0)$. What are those values ?

35 What differential equations $y' = ay + q(t)$ would be solved by $y_1(t)$ and $y_2(t)$? Jumps, ramps, corners—maybe harder than expected (math.mit.edu/dela/Pset1.4).

